

YOUR OFE 2025 REPORT

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2025 Cropping Season

Research question: Can we reduce nitrogen rates after a cover crop planted behind soybeans or dry beans?

Welcome to your 2025 farm report. This report summarizes the on-farm nitrogen rate experiment conducted in your field. Four nitrogen application rates (Control T0 = 60, T1 = 60, T2 = 50, T3 = 40 lb N/ac) were compared following a cover crop planted behind soybeans or dry beans. The goal was to determine whether a cover crop allows a reduction in nitrogen inputs without sacrificing crop nitrogen status.

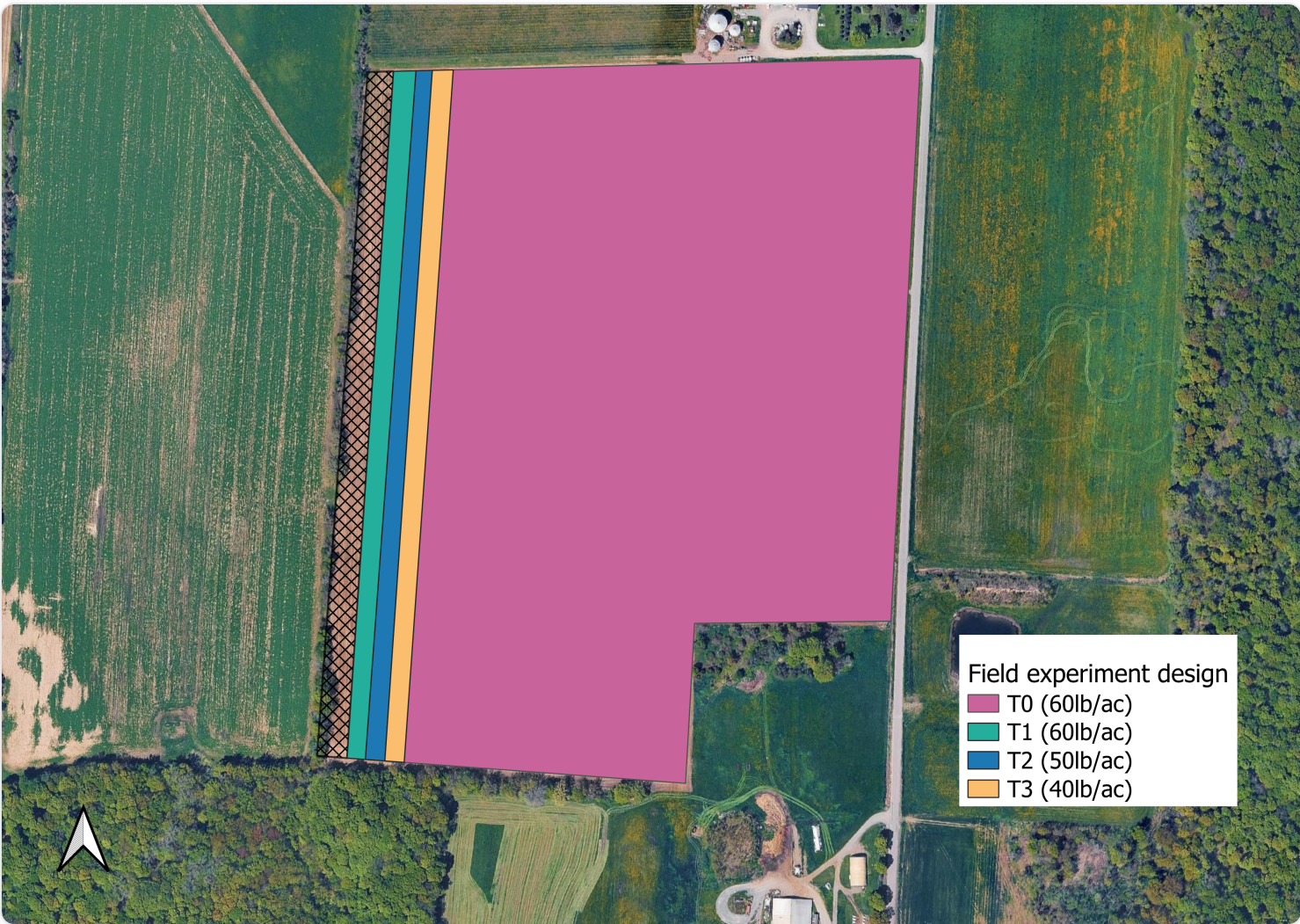


Figure 1. Treatment layout.

Take-home message: In this season, dropping the nitrogen rate from 60 down to 40–50 lb N/ac made very little difference in how much nitrogen ended up in the plant tissue. Yield was lower in the reduced-rate strips than in the control, which is the rest of the field, but the strip that got the same rate as the control (T1 = 60 lb N/ac) also yielded less, which tells us the field itself has some natural variation that’s harder to separate from the N rate effect. With only one strip per treatment, we can’t call anything a firm conclusion yet. What we can say is that the reduced rates held up reasonably well this season, and running this trial again with more strips will give us the confidence to know whether you can safely cut back on nitrogen inputs.

Results Summary

Summary of corn yield and biomass nitrogen by N-rate treatment.

Metric	n	T0 (60 lb N/ac)	T1 (60 lb N/ac)	T2 (50 lb N/ac)	T3 (40 lb N/ac)
 Corn Yield (bu/ac)	1 per strip	204 bu/ac	185 bu/ac	178 bu/ac	180 bu/ac
 Biomass %N	4 per strip	3.486 ± 0.103%	3.218 ± 0.056%	3.371 ± 0.071%	3.419 ± 0.32%
 C:N Ratio	4 per strip	13.14 ± 0.42	14.53 ± 0.24	13.78 ± 0.38	13.85 ± 1.11

n = 4 samples per treatment. T0 = Control (farmer rate, 60 lb N/ac). Results are descriptive given the small sample sizes.

Corn Yield

Corn yield was measured by strip across the four N-rate treatments (T0–T3). Each strip corresponds to one measurement. Because there is **one observation per treatment**, no statistical test is possible. Results are presented as a field-level comparison. **Keep in mind:** when two strips at the same rate give different yields, that’s the field talking, not the nitrogen.

T0 — 60 LB N/AC

204 bu/ac

T1 — 60 LB N/AC

185 bu/ac

T2 — 50 LB N/AC

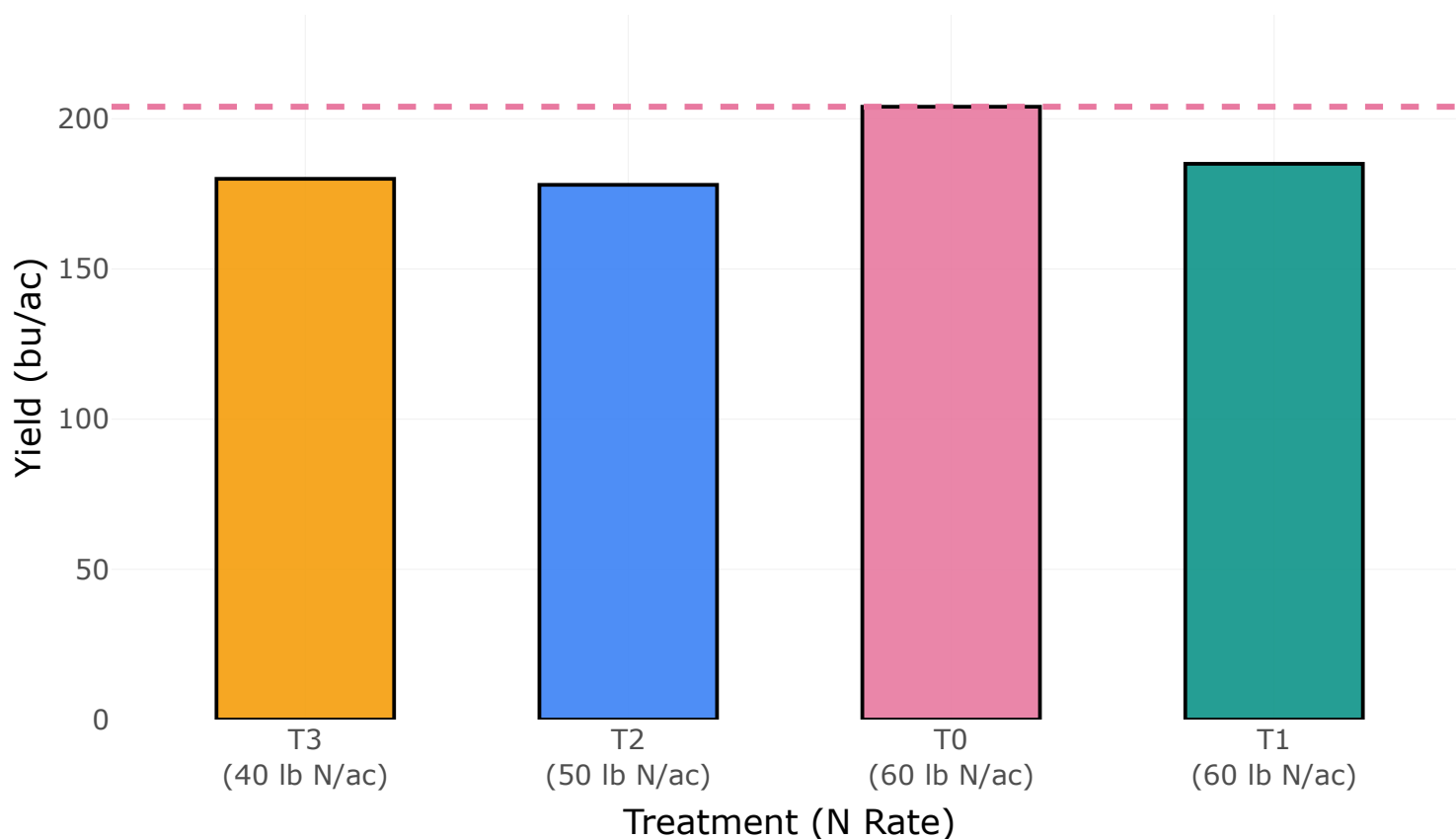
178 bu/ac

T3 — 40 LB N/AC

180 bu/ac

n = 1 per strip — field-level observation only.

Corn Yield by N-Rate Treatment



C:N in Corn Biomass

Corn biomass was collected and sent for laboratory C:N analysis. This measures the actual nitrogen concentration in plant tissue, a direct indicator of how much N the plant was able to take up at the time of sampling. **Note:** sample sizes are small (n = 4 per treatment), so results are descriptive.

Total Nitrogen in Biomass (%N)

This number tells you what percentage of the plant tissue is nitrogen. A higher %N means the plant had good access to nitrogen during the season. All four strips came in very close to each other, which is a good sign: it suggests the plants in the reduced-rate strips weren't running short on N, even with less fertilizer applied, although the 40 lb N/ac average depends heavily on one high sample.

T3 — 40 LB N/AC

3.419 %N

T2 — 50 LB N/AC

3.371 %N

T0 — 60 LB N/AC

3.486 %N

T1 — 60 LB N/AC

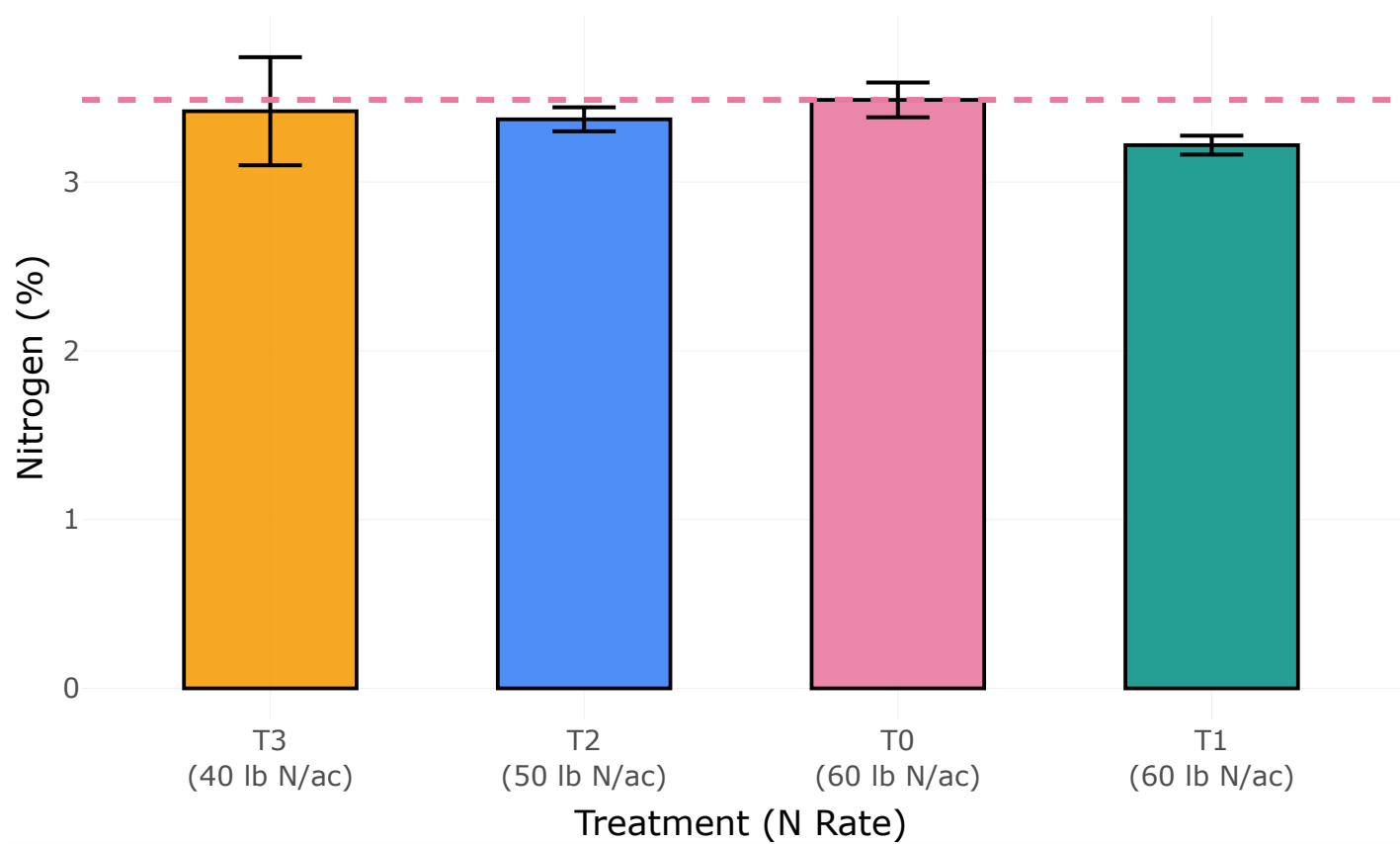
3.218 %N

n = 4 per treatment — results are descriptive only.

Bar Chart (%N)

Raw Data (%N)

Nitrogen in Corn Biomass (Mean %N $\pm$ SE)



C:N Ratio

The C:N ratio is another way of looking at the same question: how nitrogen-rich is the plant tissue? A lower C:N means the plant has more nitrogen relative to carbon, which is what you want. If the strips with less nitrogen fertilizer still show a C:N close to the control, it's a strong signal that the cover crop is helping fill that gap. In your field this season, the C:N values across all strips stayed pretty close, which is encouraging.

T3 — 40 LB N/AC

13.85 C:N ratio

T2 — 50 LB N/AC

13.78 C:N ratio

T0 — 60 LB N/AC

13.14 C:N ratio

T1 — 60 LB N/AC

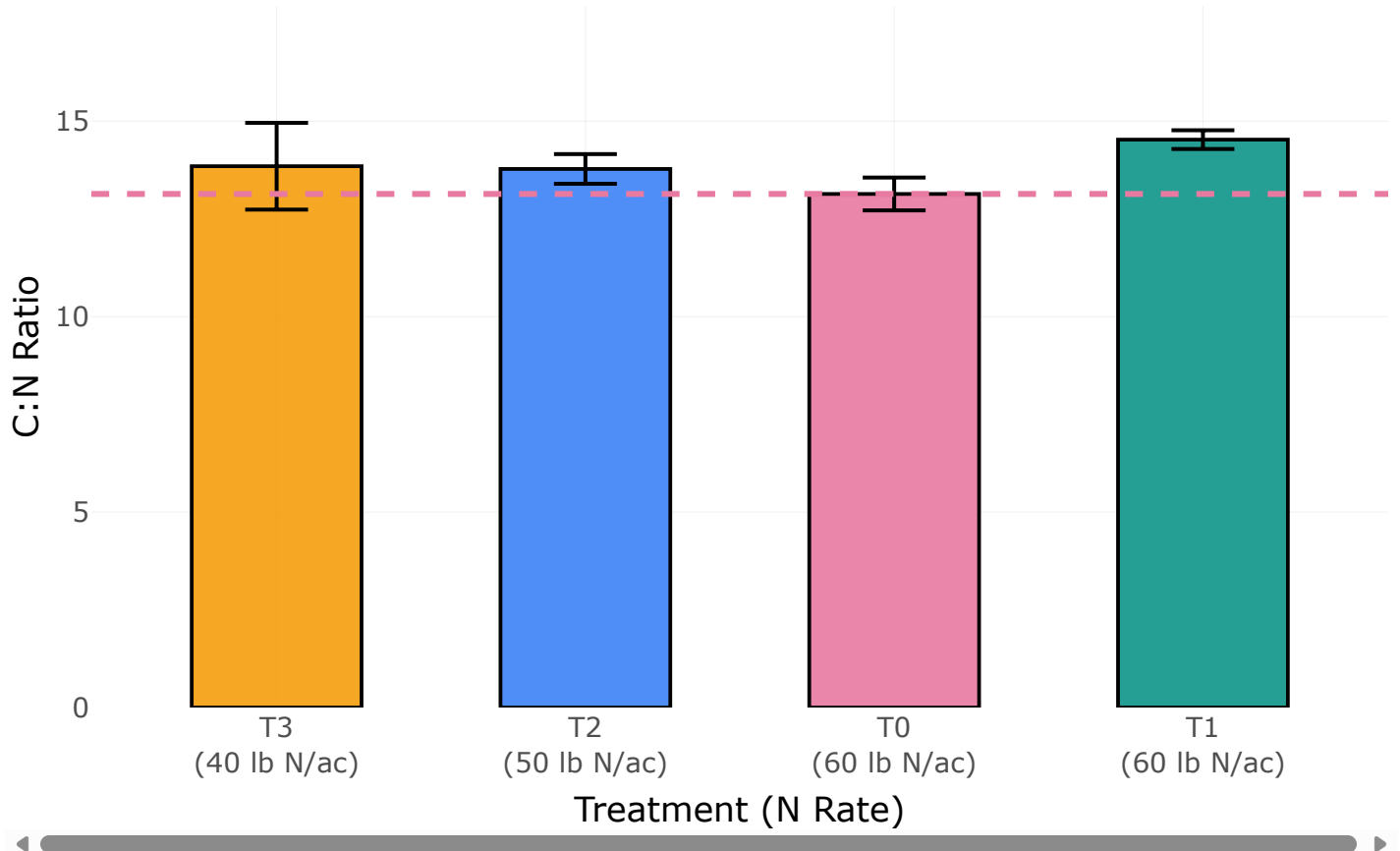
14.53 C:N ratio

Lower C:N = more nitrogen-rich tissue. n = 4 per treatment.

Bar Chart (C:N)

Raw Data (C:N)

C:N Ratio in Corn Biomass (Mean $\pm$ SE)



Soil Health Assessment

Soil health was assessed using the **Cornell Soil Health Test**. The overall quality score was **63/100 (High)**. Most physical indicators were good (Available Water Capacity: 76, Surface Hardness: 69, Subsurface Hardness: 70), although Aggregate Stability scored Low (29). The main areas for improvement were biological indicators: **Organic Matter** (score 21-Low), **Soil Respiration** (score 28-Low), and **Predicted Soil Protein** (score 32-Low). Chemical indicators were excellent, with Soil pH (100), Extractable Phosphorus (100), and Extractable Potassium (92) all in the Very High range. One flag to follow up on: **zinc came back as deficient**.

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Grower: Tom Corcoran

Agricultural Service Provider:
Alex Yore
ay327@cornell.edu

Sample ID: QCC93
Field ID: OFE 25 AFTER CC CORCORAN
Date Sampled: 07/03/2025
Coordinates: Latitude: 42.925019
Longitude: -77.92318

Measured Soil Textural Class: **loam**
Sand: **44%** - Silt: **42%** - Clay: **14%**

Group	Indicator	Value	Rating	Constraints
physical	<u>Predicted</u> Available Water Capacity	0.21	76	
physical	Surface Hardness	121	69	
physical	Subsurface Hardness	241	70	
physical	Aggregate Stability	21.1	29	
biological	Organic Matter Soil Organic Carbon: 1.36 / Total Carbon: 1.49 / Total Nitrogen: 0.14	2.3	21	
biological	<u>Predicted</u> Soil Protein	4.96	32	
biological	Soil Respiration	0.4	28	
biological	Active Carbon	509	52	
chemical	Soil pH	7.1	100	
chemical	Extractable Phosphorus	7.4	100	
chemical	Extractable Potassium	66.2	92	
chemical	Additional Nutrients Ca: 1704.9 / Mg: 192.2 / S: 5.1 Al: 4.5 / B: 0.54 / Cu: 0.16 Fe: 1.5 / Mn: 2.7 / Zn: 0.2		88	

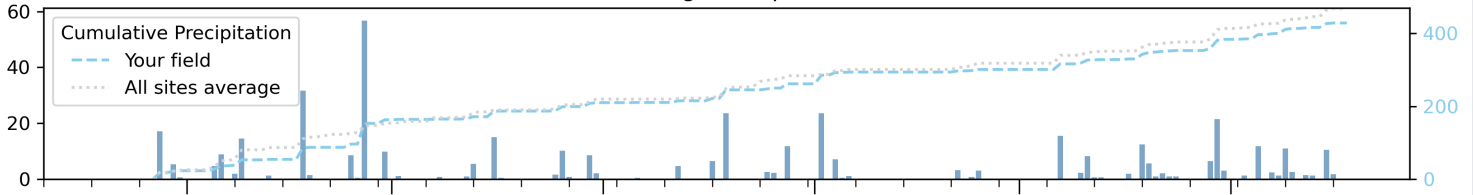
Overall Quality Score: **63** / High

[Open Full Soil Health Report \(PDF\)](#)

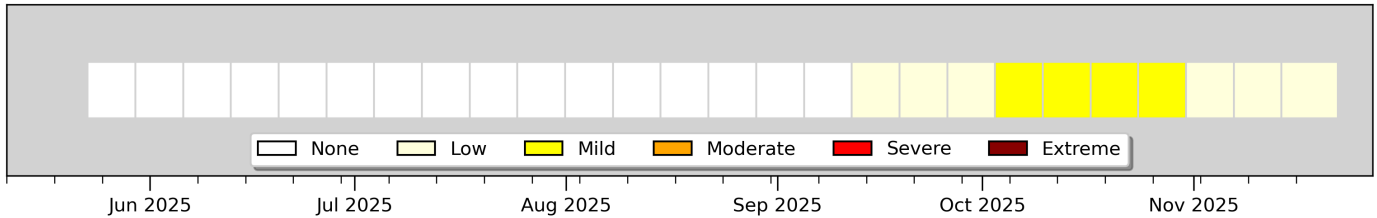
Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index from June through November 2025. Your field received strong rainfall early in the season, including a large event in late June, and tracked near the all-site average through summer. However, the drought index moved to Low in mid-September, rose to Mild for the four weeks of October, and eased back to Low in November. This late-season moisture stress is worth considering when looking at your yield results.

Local vs All Site Average Precipitation 2025 (in mm)



Weekly Drought Index



Conclusions

CORN YIELD

The Control strip (T0, 60 lb N/ac, the rest of the field) yielded **204 bu/ac**. Both reduced-rate strips were lower: T2 (50 lb N/ac) at **178 bu/ac** and T3 (40 lb N/ac) at **180 bu/ac**. Notably, T1, which received the same 60 lb N/ac as the Control T0, yielded only **185 bu/ac**. That tells us the field has some natural variability that is affecting results as much as the N rate. Compared with T1, the neighboring strip at the same 60 lb N/ac rate, the reduced-rate strips were only 5 to 7 bu/ac lower.

BIOMASS %N

Plant tissue nitrogen was nearly identical across all four strips — T0 at **3.49%**, T2 at **3.37%**, and T3 at **3.42%**. The reduced-rate strips were just barely below the control, less than 0.12 percentage points of difference. T1 came in the lowest at 3.22% despite receiving the same N rate as T0, which again points to field variability rather than a nitrogen effect. Keep in mind that the T3 average is lifted by one unusually high sample (4.37%); its other three samples were the lowest in the trial, so this result needs to be confirmed.

C:N RATIO

The control (T0) had a C:N ratio of **13.14**. T2 (13.78) and T3 (13.85) were slightly higher, meaning a touch less nitrogen-rich tissue, but still very close to the control. T1 had the highest C:N ratio of all (14.53) despite receiving the same N rate as T0, which reinforces the idea that spatial variability within the field is a bigger factor than the fertilizer rate this season.

SOIL HEALTH

The overall score was **63/100 (High)**, with excellent chemical indicators (pH, P, K all near 100). The main gaps are: Organic Matter (21), Soil Respiration (28), Predicted Soil Protein (32), and Aggregate Stability (29). As cover cropping continues, these scores should improve, potentially increasing nitrogen mineralization and the effective N credit from the cover crop. One flag to follow up on: zinc came back as deficient.